# Power BI + SQL + DWH + Python + Fabric
# Interview Questions & Easy Answers

Easy, interview-friendly reference notes

## Power BI Basics

### 1. What is Power BI used for?

Power BI is a Microsoft business intelligence tool used to connect data, clean and transform it, create reports and dashboards, and analyze business data through interactive visuals.
**Example:** We can connect Excel or SQL data to Power BI and create sales, revenue, or employee dashboards.

### 2. What steps do you follow to build a Power BI report?

First, I connect to the data source. Then I clean and transform the data using Power Query. After that, I create the data model and relationships. Then I create DAX measures, build visuals, apply filters and slicers, and finally publish the report to Power BI Service.
**Remember:** Connect → Clean → Model → DAX → Visualize → Publish

### 3. Dashboard vs Report

**Report:** Used for detailed analysis and can contain multiple pages.
**Dashboard:** Mainly used for high-level monitoring and quick business insights.

### 4. What is a dataset / semantic model?

A semantic model is the data model behind a Power BI report. It contains tables, relationships, measures, and business logic that the report uses.

### 5. What is Power Query used for?

Power Query is used to connect to data sources and clean and transform the data before loading it into Power BI. Examples: remove duplicates, handle nulls, change data types, merge tables, append tables, and rename columns.

### 6. Measures vs Calculated Columns

**Calculated Column:** Calculated row by row and stored in the model.
**Measure:** Calculated when it is used in a visual and depends on the current filter context.

### 7. Import vs DirectQuery

**Import:** Data is loaded into Power BI memory, so reports are generally faster.
**DirectQuery:** Data stays in the source database and Power BI sends queries to the source when data is needed.
**Remember:** Import = data comes to Power BI; DirectQuery = Power BI goes to the data source.

### 8. What is a slicer?

A slicer is a visual filter that allows users to select values and filter other visuals on the report page. Example: selecting 2025 from a Year slicer shows only 2025 data.

### 9. How do you publish a report?

I create the report in Power BI Desktop, save it, and then click Publish. I select the required workspace in Power BI Service.

### 10. What is refresh? Why does it fail?

Refresh means updating Power BI data with the latest data from the source. It can fail because of database connection problems, incorrect credentials, gateway issues, changed table or column names, or an unavailable source.

### 11. Name 3 visuals you used and why

I have used bar charts, line charts, and cards. Bar chart → compare categories. Line chart → show trends over time. Card → show important KPIs like Total Sales or Profit.

## DAX Interview Questions

### 12. SUM vs CALCULATE

SUM simply adds values from a column. CALCULATE changes or applies the filter context before calculating an expression.

### 13. Row Context vs Filter Context

**Row context:** Power BI is looking at one row at a time.
**Filter context:** The calculation is affected by filters from slicers, visuals, or DAX. Example: selecting 2025 from a Year slicer makes a measure calculate only for 2025.

### 14. SUM vs SUMX

SUM adds values from one column. SUMX goes row by row, performs a calculation, and then adds the results. Use SUM when the required values already exist in a column; use SUMX when you need a calculation for each row before adding the results.

### 15. What is context transition?

Context transition happens when CALCULATE converts the current row context into filter context. For example, when CALCULATE is used inside an iterator or calculated column, it can convert the current row into a filter.

### 16. YTD Sales

YTD means Year To Date. It calculates sales from the beginning of the year up to the current date.
```
YTD Sales = TOTALYTD([Total Sales], 'Date'[Date])
```

### 17. MoM Growth %

MoM means Month over Month. It compares the current month's value with the previous month's value.
```
MoM Growth % = DIVIDE([Total Sales] - [Previous Month Sales], [Previous Month Sales])
```

### 18. Running Total

Running total shows the accumulated value up to the current date.

### 19. Rolling 7 Days

A rolling 7-day calculation shows the total for the current day and the previous six days.

### 20. ALL / ALLEXCEPT / ALLSELECTED

**ALL:** removes filters.
**ALLEXCEPT:** removes filters except the specified column(s).
**ALLSELECTED:** respects the user's selections while handling visual-level filtering.

### 21. How do you debug wrong numbers in DAX?

First, I check the source data. Then I check relationships between tables, filters and slicers, data types, and the DAX formula. I also create small test measures to identify where the number is changing.

## SQL Interview

### 22. Write a query using JOIN

JOIN is used to combine data from two or more tables using a related column.

```
SELECT e.name, d.department
FROM employees e
INNER JOIN departments d
ON e.dept_id = d.dept_id;
```

### 23. GROUP BY + HAVING

GROUP BY creates groups, and HAVING filters those groups.

```
SELECT department, COUNT(*) AS employee_count
FROM employees
GROUP BY department
HAVING COUNT(*) > 2;
```

### 24. Find duplicate records

```
SELECT name, COUNT(*)
FROM employees
GROUP BY name
HAVING COUNT(*) > 1;
```

### 25. ROW_NUMBER

ROW_NUMBER gives a unique sequential number to each row based on the specified ordering.

```
SELECT name, salary,
ROW_NUMBER() OVER(ORDER BY salary DESC) AS rn
FROM employees;
```

### 26. Primary Key vs Foreign Key

Primary Key uniquely identifies each record and cannot contain NULL values. Foreign Key creates a relationship between tables by referring to a key in another table.

### 27. WHERE vs HAVING

WHERE filters rows before grouping. HAVING filters groups after GROUP BY.

### 28. INNER JOIN vs LEFT JOIN

INNER JOIN returns only matching records from both tables. LEFT JOIN returns all records from the left table and matching records from the right table.

### 29. Find top 5 products by revenue

```
SELECT product_id, SUM(revenue) AS total_revenue
FROM sales
GROUP BY product_id
ORDER BY total_revenue DESC
LIMIT 5;
```

### 30. Count customers per city

```
SELECT city, COUNT(*) AS customer_count
FROM customers
GROUP BY city;
```

### 31. What is a subquery?

A subquery is a query written inside another query. Example: find employees whose salary is greater than the average salary.

### 32. What is indexing?

An index helps the database find data faster, similar to an index in a book. Indexes can improve SELECT performance but can add overhead to INSERT, UPDATE, and DELETE.

### 33. Normalization vs Denormalization

Normalization divides data into multiple related tables to reduce duplication. Denormalization combines or duplicates some data to make querying and reporting faster or simpler.

### 34. How do you handle NULLs?

First I understand why the value is NULL. Depending on the requirement, I can use IS NULL, IS NOT NULL, COALESCE, or replace NULL with an appropriate default value.

### 35. RANK vs DENSE_RANK

For salaries 70,000, 70,000, 60,000: RANK = 1, 1, 3. DENSE_RANK = 1, 1, 2. RANK skips numbers after a tie; DENSE_RANK does not.

### 36. UNION vs UNION ALL

UNION combines results and removes duplicates. UNION ALL combines results but keeps duplicates.

## Data Modeling

### 37. What is a Fact Table?

A fact table stores measurable business events or transactions. Example: Sales Fact with Date, Product, Customer, Quantity, and Sales.

### 38. What is a Dimension Table?

A dimension table contains descriptive information used to analyze facts. Examples: Customer, Product, Date, Employee, and Location.

### 39. What is Star Schema?

Star schema is a data model where a central fact table is connected to multiple dimension tables.

### 40. Why do we prefer Star Schema?

Star schema is simple to understand, provides better performance, makes relationships easier to manage, and works well with Power BI.

### 41. What is a Date Table?

A date table is a dedicated table containing dates and attributes such as year, month, quarter, and day. It helps create YTD, MTD, QTD, Previous Year, MoM, and other time-based analysis.

# ETL + Data Warehouse

## 42. What is ETL?

ETL stands for Extract, Transform, Load. Extract = get data from sources. Transform = clean and transform data. Load = load data into the data warehouse.

## 43. Full Load vs Incremental Load

Full Load loads all data every time. Incremental Load loads only new or changed data. If 10 million records exist and only 10,000 changed today, incremental loading loads the changed records instead of all 10 million.

## 44. What is data quality?

Data quality means checking whether data is accurate, complete, consistent, valid, and reliable. Checks include NULLs, duplicates, data types, primary keys, foreign keys, ranges, and record counts.

## 45. What is a surrogate key?

A surrogate key is an artificial or system-generated key used to uniquely identify records in a dimension table.

## 46. What is SCD?

SCD stands for Slowly Changing Dimension. It is used to manage changes in dimension data over time.

## 47. SCD Type 1 vs Type 2

Type 1 updates the old value and does not keep history. Type 2 keeps history by creating a new record. Type 1 overwrites; Type 2 maintains historical records.

# Python / API

## 48. How do you read a CSV?

```
import pandas as pd
df = pd.read_csv("sales.csv")
```

## 49. How do you handle missing values?

First I identify missing values. Depending on the business requirement, I can remove them or replace them with an appropriate value such as 0, mean, median, or another meaningful value.

## 50. List vs Dictionary

A list stores multiple values in an ordered collection. A dictionary stores data as key-value pairs.

## 51. What is groupby in Pandas?

Groupby is used to group data based on one or more columns and perform calculations such as sum, count, or average.

```
df.groupby("department")["salary"].sum()
```

## 52. What is an API?

API stands for Application Programming Interface. It allows two applications or systems to communicate with each other.

## 53. GET vs POST

GET is used to retrieve data. POST is used to send or create data.

### 54. What is pagination?

Pagination means getting a large amount of API data in smaller parts or pages instead of getting everything at once.

### 55. What is a rate limit?

A rate limit controls how many API requests we can make within a certain period.

### 56. Pandas vs PySpark

Pandas is generally suitable for smaller datasets that can fit into memory on one machine. PySpark is better for very large datasets because it can distribute processing across multiple machines.

### 57. Spark Transformation vs Action

Transformation creates a new dataset or operation plan but does not immediately execute it. Examples: filter(), select(), groupBy(). Action triggers execution and produces a result. Examples: count(), show(), collect().

## Microsoft Fabric

### 58. What is Microsoft Fabric?

Microsoft Fabric is an end-to-end analytics platform from Microsoft. It brings together data engineering, data integration, data warehousing, data science, real-time analytics, and Power BI in one platform. Power BI mainly focuses on analytics and reporting, while Fabric provides a broader platform for the complete data and analytics lifecycle.

### 59. What is OneLake?

OneLake is the centralized data lake for Microsoft Fabric. It provides a single place to store and access organizational data across different Fabric workloads.

### 60. Lakehouse vs Data Warehouse in Fabric

A Lakehouse combines features of a data lake and a data warehouse and is useful for flexible data storage and engineering workloads. A Fabric Warehouse is mainly designed for structured relational data and SQL-based analytics.

### 61. What is a Fabric Pipeline?

A Fabric pipeline is used to automate and orchestrate data movement and data processing tasks. We can schedule pipelines to extract data from sources, transform it, and load it into the required destination.

## Communication Questions

### 62. Explain your project in 2 minutes

I worked on a data analytics project where I collected data from different sources such as SQL and Excel. I used SQL for data extraction and transformation and Power Query for data cleaning. I created a star schema with fact and dimension tables and developed DAX measures such as Total Sales, YTD Sales, and Growth %. Then I created Power BI dashboards using cards, charts, and slicers. Finally, I published the report to Power BI Service and configured refresh. The main goal of the project was to provide business users with an interactive dashboard to analyze performance and make better decisions.

### 63. What if the report shows wrong numbers?

First, I don't immediately change the DAX. I check the source data, relationships, filters, data types, and then the DAX calculation. I compare the Power BI result with the source SQL result to identify where the difference is coming from.

### 64. What challenges did you face?

One challenge was data quality issues such as NULL values, duplicates, and inconsistent data. I handled them during the transformation stage using SQL and Power Query. Another challenge was incorrect numbers in some visuals, which I resolved by checking relationships, filters, and DAX measures.

## Priority for Interview Preparation

### Priority 1 — Must Know

What is Power BI?; Power Query; Report vs Dashboard; Import vs DirectQuery; Measure vs Calculated Column; SUM vs SUMX; CALCULATE; Row Context vs Filter Context; YTD; SQL JOIN; WHERE vs HAVING; GROUP BY; Primary Key / Foreign Key; ROW_NUMBER; RANK vs DENSE_RANK; Fact vs Dimension; Star Schema; ETL; Full Load vs Incremental Load; SCD Type 1 vs Type 2.

### Priority 2

ALL / ALLEXCEPT / ALLSELECTED; Context Transition; Running Total; MoM Growth; Duplicate records; Subquery; Indexing; Normalization; Data quality checks; Surrogate key.

### Priority 3 — JD/Fabric

Microsoft Fabric; OneLake; Lakehouse; Fabric Warehouse; Fabric Pipeline; Pandas; API; GET vs POST; Pagination; PySpark.

### Final Interview Tip

Do not try to memorize all answers word-for-word. Understand the concept and explain it naturally. A useful project flow to remember is: Source → SQL → ETL → Model → DAX → Power BI → Publish.